

# Karsa HV board manual

## Introduction

Karsa Hv board is designed to provide one to three bipolar high voltage sources. Each channel consists of a pair of HVDCDC-converters and HV relays for switching between positive and negative outputs. Each pair can be populated with XP Power CB101(N)-sources for +-10kV use or CA05(N) modules for +-500V use.

HVDCDC module control source can be selected by a jumper from three sources: onboard potentiometers, external analog voltage or DA-converter as controlled by a ST microcontroller. Relay control is implemented by electronics directly. Inputs for relay controls are input analog signals of both HV converters. It is set so that output relay is activated only if corresponding converter is active and the pair is not (XOR logic).

Converter modules provide current and voltage monitor analog signals. These are divided by two over resistor bridge and then read out by STM's internal ADC. All possible DC converters map their signal 0...5V range to 0...max.

Communication to other systems is done via CANopen inspired interface. The board will only respond to limited SDO messages, which can be used to inquire and set voltages. CAN bus speed is hardcoded to 1Mbit/s. CAN address can be set by DIP switch on board.

Save/load parameters command stores or retrieves values for indices 0x2000-0x2003. If no parameters are stored on bootup, following default parameters are loaded. (Reset value of 0xFFFF on 0x2000 sub 1 loads the defaults). Otherwise, parameters are automatically loaded from flash. Note that storage doesn't use any load balancing and flash memory is susceptible to wear if erased excessively (~1000 cycles).

|   | 0x2000 | 0x2001 | 0x2002 | 0x2003 |
|---|--------|--------|--------|--------|
| 1 | 1000   | 10     | 10000  | 100    |
| 2 | 100    | 100    | 500    | 2000   |
| 3 | 1      | 1      | 1      | 1      |

## CAN objects

| Index  | Command                                      | Subindex | read/<br>write | Value                                                                                                    | Data<br>length<br>(bytes) |
|--------|----------------------------------------------|----------|----------------|----------------------------------------------------------------------------------------------------------|---------------------------|
| 0x1001 | Read Error Register                          | X        | ro             | four byte error register *                                                                               | 4                         |
| 0x1002 | Read status                                  | X        | ro             | four byte status register *                                                                              | 4                         |
| 0x1009 | Hardware version                             | X        | ro             | HW version                                                                                               | 1                         |
| 0x100A | Firmware version                             | X        | ro             | FW version                                                                                               | 1                         |
| 0x1010 | Load / store parameters                      |          | 1 wo           | Store parameters to flash with data values ['s','a','v','e'] or read with ['l','o','a','d'] <sup>1</sup> | 4                         |
| 0x1017 | Heartbeat interval                           | X        | rw             | Heartbeat interval in 10ms, 0 to disable                                                                 | 2                         |
| 0x2000 | Voltage reading scale factor <sup>2</sup>    | 1-maxCH  | rw             | Unit of voltage monitor and set values in mV                                                             | 2                         |
| 0x2001 | Current reading scale factor <sup>2</sup>    | 1-maxCH  | rw             | Unit of current monitor values in nA                                                                     | 2                         |
| 0x2002 | Channel max voltage <sup>2</sup>             | 1-maxCH  | rw             | Max voltage in V                                                                                         | 2                         |
| 0x2003 | Channel max current <sup>2</sup>             | 1-maxCH  | rw             | Max current in $\mu$ A                                                                                   | 2                         |
| 0x7130 | Read last voltage monitor value <sup>2</sup> | 1-maxCH  | ro             | Channel output voltage in units specified by 0x2000                                                      | 2                         |
| 0x7131 | Read last current monitor value <sup>2</sup> | 1-maxCH  | ro             | Channel output current in units specified by 0x2001                                                      | 2                         |
| 0x7300 | Voltage setpoint <sup>2</sup>                | 1-maxCH  | rw             | Channel setpoint in units of 0x2000                                                                      | 2                         |

X = don't care; \* = not implemented; <sup>1</sup> = excessive use will wear flash down; <sup>2</sup> = reading subindex 0 will return number of channels

Old table, don't use

| Index  | Command                      | Subindex | Accepted commands (byte 0) | Value                                                                                                    | Data length (bytes) |
|--------|------------------------------|----------|----------------------------|----------------------------------------------------------------------------------------------------------|---------------------|
| 0x1001 | Read Error Register          | X        | 0x40                       | four byte error register *                                                                               | 4                   |
| 0x1002 | Read status                  | X        | 0x40                       | four byte status register *                                                                              | 4                   |
| 0x1009 | Hardware version             | X        | 0x40                       | HW version                                                                                               | 1                   |
| 0x100A | Firmware version             | X        | 0x40                       | FW version                                                                                               | 1                   |
|        |                              |          |                            | Store parameters to flash with data values ['s','a','v','e'] or read with ['l','o','a','d'] <sup>1</sup> |                     |
| 0x1010 | Load / store parameters      |          | 1 0x22                     |                                                                                                          | 4                   |
| 0x1017 | Heartbeat interval           | X        | 0x40, 0x22                 | Heartbeat interval in ms/10                                                                              | 2                   |
|        | Current reading scale factor |          | 0 0x40                     | Number of channels                                                                                       | 1                   |
|        |                              | 1-maxCH  | 0x40, 0x22                 | Unit of current monitor values in nA                                                                     | 2                   |
| 0x6442 | Voltage scale factor         |          | 0 0x40                     | Number of channels                                                                                       | 1                   |
|        |                              | 1-maxCH  | 0x40, 0x22                 | Unit of voltage monitor and set values in mV                                                             | 2                   |
| 0x7130 | Read last monitor value      |          | 0 0x40                     | Number of channels                                                                                       | 1                   |
|        |                              | 1-maxCH  | 0x40                       | Channel output voltage in units specified by 0x6442                                                      | 2                   |
| 0x7300 | Set value                    |          | 0 0x40                     | Number of channels                                                                                       | 1                   |
|        |                              | 1-maxCH  | 0x40, 0x22                 | Channel setpoint (rw)                                                                                    | 2                   |
|        |                              |          |                            |                                                                                                          |                     |

X= don't care; \* = not implemented; <sup>1</sup> = excessive use will wear flash down